



Tensile behavior of 2D-C/SiC composites at elevated temperatures: Experiment and modeling



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ABSTRACT

In this paper, on-axis tensile behavior of a coated 2D-C/SiC composite at elevated temperatures was studied experimentally and theoretically. The measured data reveals that the tensile modulus and strength increase continuously with increasing temperature till 1273 K. Contrarily, the failure strains decrease sharply at high temperatures than the counterpart at room temperature, manifesting the significant influence of thermal residual stresses (TRS) on mechanical behavior of C/SiC composites. Simulation of stress-strain response is based on a two-scale analytical model, in which the plain-weave element is idealized as a cross-ply laminate and its macroscopic mechanical parameters are evaluated by shear-lag approach. The primary calculation was concentrated on TRS of the composite. And, a new crack evolution model was introduced to describe the stochastic cracking process. The total strain response including residual strain and elastic strain from the loading-unloading-reloading conception was finally formulated through micromechanical analysis involving the influence of TRS on matrix cracking and interface debonding. Additionally, a strength model was developed for plain-weave structures by using shear-lag theory, statistical theory and rule of mixture. Both of the proposed constitutive and strength models can give accurate predictions for 2D-C/SiC composites at elevated temperatures.

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1. Introduction

Carbon fibers as well as silicon carbide matrix are known to have excellent high temperature performance [1,2]. Meanwhile, their compounds, i.e., C/SiC composites, inherit the outstanding properties of the fibers and matrix while overcoming the brittleness of monolithic ceramic. Such composites are promising for application as high temperature structural materials due to their high specific stiffness and strength, especially the high temperature performance over conventional superalloys in order that little or no cooling is required [3,4].

Prior to the application of C/SiC materials in hot structures, characterization of their mechanical behavior in high temperature environments is required for component design. Many previous efforts have been devoted to experimental and analytical investigations on the thermo-mechanical properties of C/SiC composites [5–19], with emphasis on the temperature and oxygen effect, microstructural degradation law and mechanical evolution mechanisms. In these studies, the high temperature in-plane and interlaminar shear strength stability of 2D-C/SiC composites are

presented, without tensile performance. Furthermore, only a few reports have been found toward the high temperature stress-strain behaviors of C/SiC composites under various loading cases [5,6,12,18] due likely to the technical difficulty in precisely measuring strain.

In order to characterize the high temperature behavior of ceramic composites, it is necessary to determine the thermal stresses. TRS [20–22] are known to have remarkable influence upon the macroscopic mechanical behavior of C/SiC composites, which determine the stress states of the constituents (compressive or tensile stresses) in the composites and affect damage evolution such as matrix cracking and interfacial debonding. More importantly, such damages in turn release the internal TRS. Consequently, estimation of thermal stresses is a very important issue for the development and application of advanced C/SiC materials. Once the significant thermal stresses are completely determined, the damage evolutionary mechanisms under different loading conditions can be understood and the potential improvement of mechanical properties of these materials can be made.

In view that the high temperature constitutive relationships of ceramic matrix composites have rarely been investigated, the present study concentrates primarily on the nonlinear tensile behavior of a 2D-C/SiC composite at elevated temperatures. The stress-strain response and rupture strength are investigated

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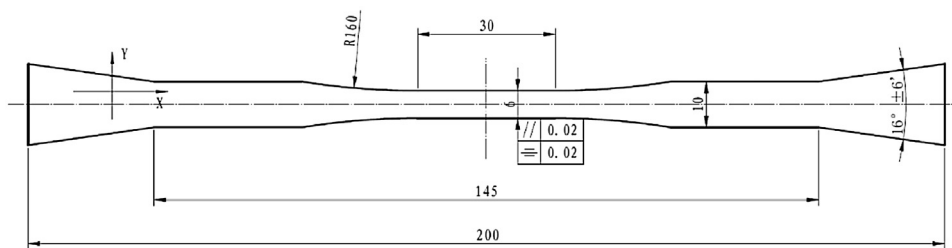


Fig. 1. The shape and basic dimensions of the specimen.

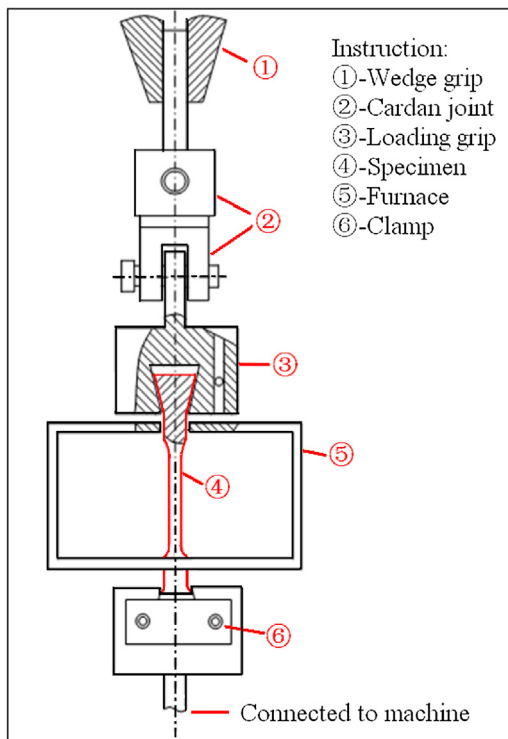


Fig. 2. The designed device for high temperature tensile tests.

experimentally and analytically. And, a two-scale theoretical model is formulated to account for the influence of thermal stresses on the damage mechanisms of 2D-C/SiC composites.

2. Tensile experiment

2.1. Material and test

The material used in this study is plain-weave C/SiC composite fabricated by CVI technique. The fiber volume fraction is about 40%. Before deposition of SiC matrix, a thin pyrolytic carbon layer of 1.5–2.0 μm in thickness was deposited on the surface of the fibers as the interfacial layer. The dog-bone specimens were machined from a composite panel, and then coated with SiC by I-CVI process. The final density of the coupons is about 2.0 g/cm^3 . The shape and basic dimension of the specimen are shown in Fig. 1.

The uniaxial tensile experiments were performed in air at room temperature, 973 K, 1273 K and 1473 K on a mechanical testing machine (CSS-DNS-100, Changchun, China) with a 100 kN load cell under displacement control at a crosshead speed of 0.3 mm/min. Strains were continuously recorded by a clip-on extensometer with a gauge length of 25 mm (Model: Epsilon 3549, USA). The experimental devices for high temperature tensile tests are schematically

presented in Fig. 2. The heating rate of the furnace is about 20 $^{\circ}\text{C}/\text{min}$.

2.2. Results and discussion

2.2.1. Stress-strain curve

Fig. 3(a) shows the typical on-axis tensile behavior of the 2D-C/SiC composite at room temperature. It is observed that the stress-strain response of the material is dramatically nonlinear. And, nonlinearity starts almost from the onset of loading. The initial modulus is about 128.7 GPa obtained from the beginning of the stress-strain curve. Besides, the measured failure strain and strength are 0.55% and 225.8 MPa, respectively.

The high temperature tensile features of the 2D-C/SiC composite are characterized in Fig. 3(b)–(d), respectively. It is clearly seen that the trace of each tested curve is approximately parabolic. Furthermore, the proportional limit stress is increased with elevated temperature (see Table 1). Identical tendency is found in the strength and stiffness data till 1273 K, and then a decrease is observed at 1473 K. Contrarily, the failure strains at 973 K, 1273 K and 1473 K are surprisingly much smaller than that at room temperature, revealing different damage evolution mechanisms at varied temperatures. These phenomena will be theoretically demonstrated in Sections 3 and 4.

2.2.2. Damage mechanism

The damage patterns of tensile tests at elevated temperatures are schematically shown in Fig. 4. It is seen from the polished section (a) and (b) that the edges of specimen were obviously oxidized during heating up even though the SiC coatings were deposited on the surfaces. Meanwhile, with higher test temperature the heating up time is longer and the oxidation recession is worse. Slight internal erosion is also observed for samples exposed at or beyond 1273 K except for the near-surface oxidation. Toward crack evolution, Fig. 4(c) shows a crack propagating from transverse fiber bundle to the longitudinal and further inducing oxidation. Fiber-matrix interface debonding is also observed as presented in Fig. 4(d).

The main involved damage mechanisms accounting for the remarkable influence of TRS on tensile features of 2D-C/SiC composites are matrix cracking and interfacial debonding. Actually, matrix cracks already exist in the as-fabricated material due to large thermal expansion mismatch between the constituents, i.e. fiber and matrix. Upon loading, new cracks will generate in the transverse and longitudinal bundles when the external load exceeds so-called cracking stress, which is apparently related to the proportional limit stress and is obviously elevated when the test temperature gradually increases. Furthermore, load bearing status of the constituents is also strongly influenced by TRS. Once this internal stress can be diminished, the mechanical properties of 2D-C/SiC composites would increase evidently. These damage mechanisms provide the basis for developing micromechanical model.

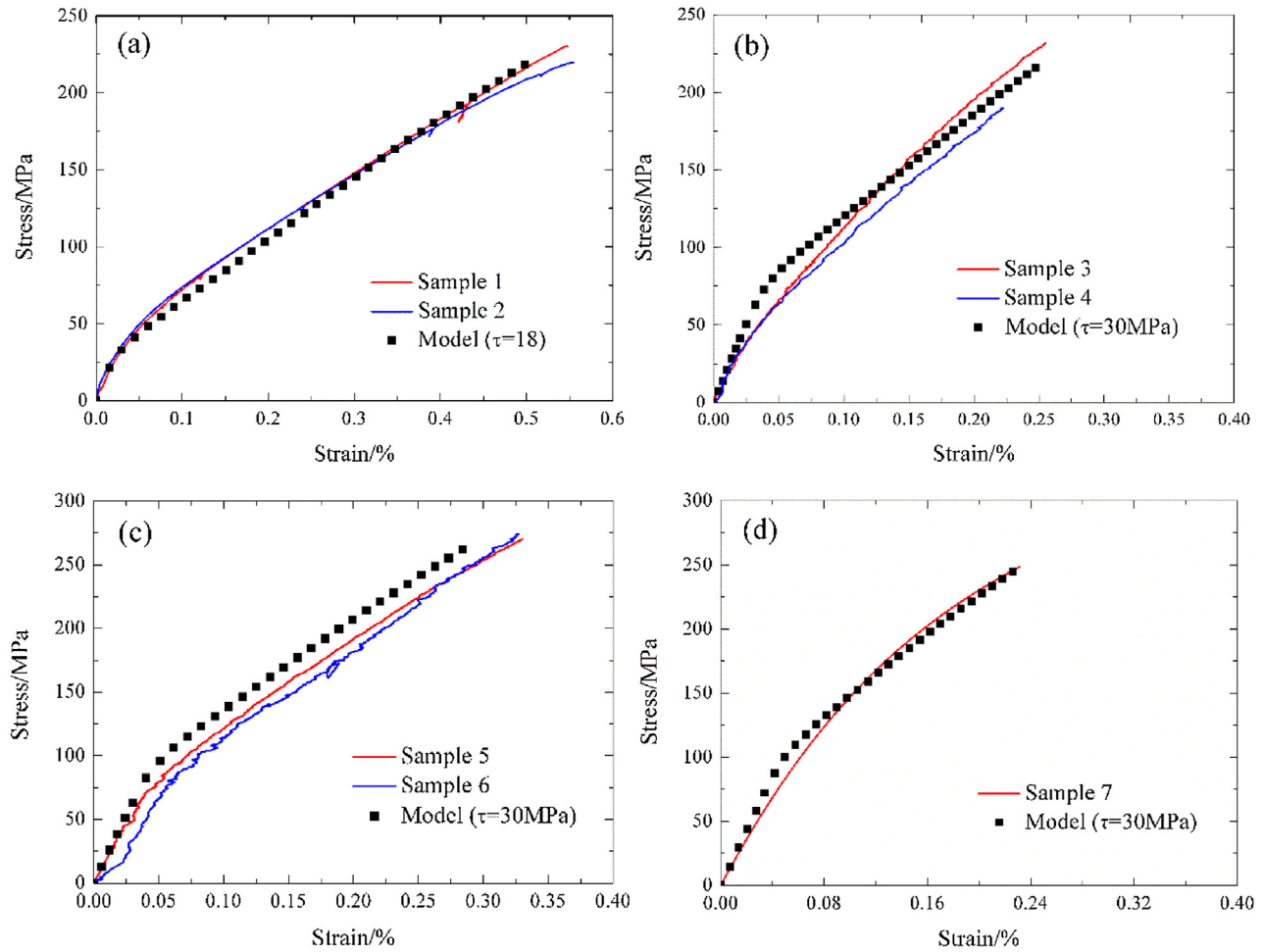


Fig. 3. Tensile behavior of 2D-C/SiC composite at elevated temperatures: (a) room temperature, (b) 973 K, (c) 1273 K, (d) 1473 K.

Table 1

Tensile experimental results at elevated temperatures.

Temperature (K)	Material density (g/cm ³)	Proportional limit stress (MPa)	Modulus (GPa)	Strength (MPa)	Failure strain (%)	Number of tests
300	2.0	30	128.7 ± 1.8	225.8 ± 8.50	0.55 ± 0.5%	2
973	2.0	50	152.3 ± 4.7	240.5 ± 13.4	0.24 ± 1.9%	5
1273	2.0	80	172.7 ± 7.0	268.2 ± 19.3	0.32 ± 2.9%	6
1473	2.0	100	169.1 ± 2.6	240.9 ± 10.5	0.25 ± 2.1%	2

3. Tensile model formulation

3.1. TRS of 2D-C/SiC composite

The real morphology of the fiber bundles in 2D-C/SiC composite is presented in Fig. 5(a). It is observed that the curvature of the fiber tow is relatively small. In order to simplify the analysis, the plain-weave element is idealized to a cross-ply one (Fig. 5(b)), and then a mini-composite model shown in Fig. 5(c) is used to evaluate the mechanical properties of a single layer. In the shear-lag model, L is the instantaneous matrix crack spacing at applied stress level σ , L_d is the interface debonding length, u is the crack opening displacement.

The stress field in the cross-ply material held at an initial temperature T_0 comprises TRS at the laminate level due to differential expansion of 0° and 90° plies, and TRS at the ply level inside the 0° plies due to differences in coefficient of thermal expansion (CTE) between fibers and matrix. Once the imposed stress exceeds a

critical level, new cracks are supposed to initiate in 90° layers and rapidly spread to 0° layers as large tensile TRS exists in the matrix. Subsequently, matrix cracks are present in the longitudinal bundle and simultaneously propagate across the section, followed by fiber-matrix interface debonding.

The two-scale mechanical model mentioned above is applied to correlate the micro- and macroscopic damage evolutions of the 2D-C/SiC composite. In order to describe the influence of TRS on mechanical behavior of the material at elevated temperatures, almost all the material parameters presented in the following are considered as temperature dependent.

3.1.1. Interlaminar TRS

The longitudinal modulus E_L and transverse modulus E_T of a single layer at arbitrary temperature can be approximated by [23]

$$E_L = V_f E_f^L + V_m E_m \quad (1)$$

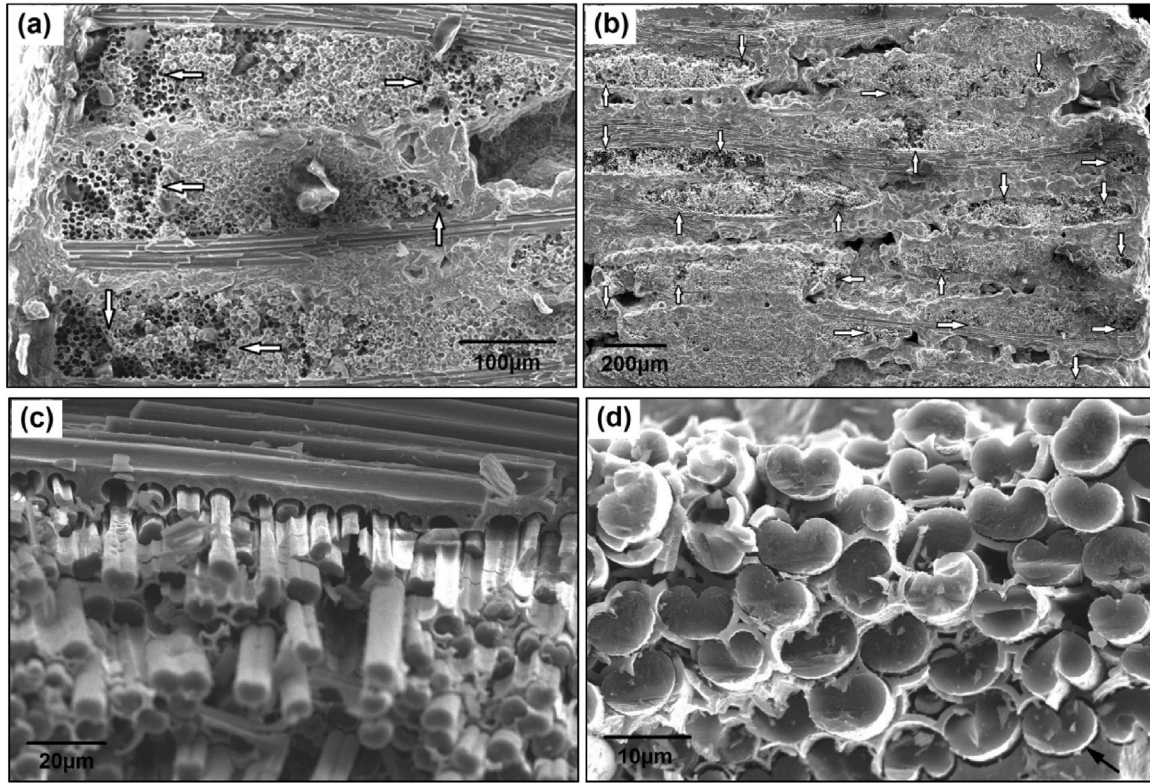


Fig. 4. Damage mechanisms of tensile tests: (a) near-surface oxidation, (b) internal oxidation, (c) crack propagating from transverse bundle to the longitudinal, (d) fiber-matrix interface debonding.

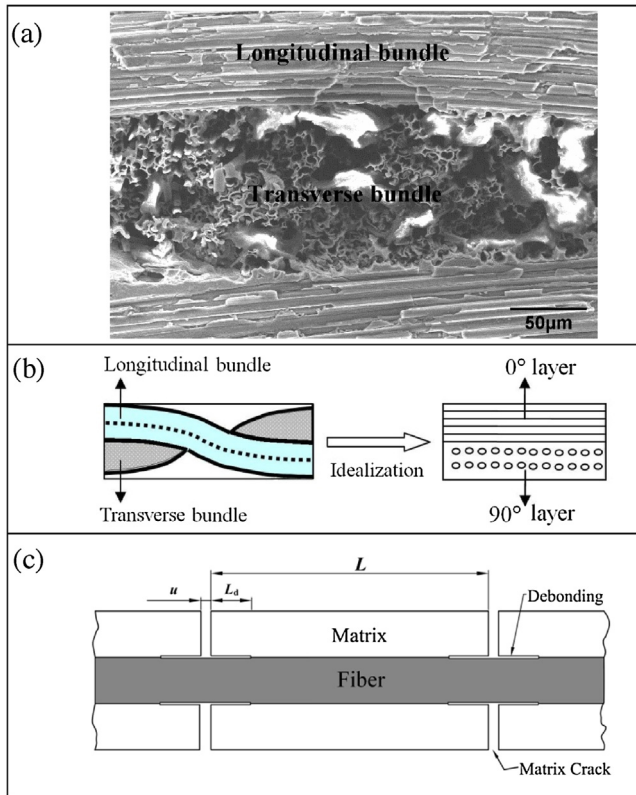


Fig. 5. Mechanical model construction: (a) morphology of the fiber bundle in the 2D-C/SiC composite, (b) idealization of the plain-woven structure to a cross-ply material, (c) the mini-composite model to evaluate the properties of a single layer.

$$E_T = \frac{\sqrt{V_f}}{\sqrt{V_f/E_f^T + (1 - \sqrt{V_f})(1 - \nu^2)/E_m}} + (1 - \sqrt{V_f}) \frac{E_m}{1 - \nu^2} \quad (2)$$

where E_f^L and E_f^T are the axial and radial moduli of the fiber, respectively; E_m is the modulus of the matrix; ν is Poisson's ratio of the matrix and fiber (assumed to be the same); V_f and V_m the volume fraction of fiber and matrix, respectively.

The average longitudinal and transverse thermal expansion coefficients of a single layer, related to the temperature, can be calculated by

$$\alpha_1 = [\alpha_f^L V_f E_f^L + \alpha_m V_m E_m] / E_L \quad (3)$$

$$\alpha_2 = \alpha_f^T V_f + \nu \alpha_f^L V_f + (1 + \nu) V_m \alpha_m - \nu \alpha_1 \quad (4)$$

where α_f^L and α_f^T are the specific axial and radial CTE of the fiber, respectively; while α_m is the specific CTE of the matrix.

In most cases, the value of α_1 is different from α_2 . Subsequently, when cooling down the material from processing temperature T_0 to test temperature T , a level of residual stress may arise between the 0° layer and 90° layer, that is

$$\sigma_L^R = \Omega_1 E_L E_T / (E_L + E_T) \quad (5)$$

Ω_1 is the misfit strain and can be estimated by

$$\Omega_1 = \alpha_2(T) - \alpha_2(T_0) - \alpha_1(T) + \alpha_1(T_0) \quad (6)$$

If the 0° bundle possesses compressive residual stress, it would not incline to suppress the tensile deformation of itself. Then, an apparent thermal stress can be defined as

$$\sigma_{th}^{90} = -\sigma_L^R / 2 \quad (7)$$

which will be used to predict the cracking behavior of the 90° layer in Section 3.2.2. The associated residual stress introduced into the matrix of the 0° layer is

$$\sigma_m^{R1} = \sigma_L^R E_m / E_L \quad (8)$$

Eq. (8) is based on the equal strain assumption.

3.1.2. Intralaminar TRS

In order to investigate the damage behavior of the composite, the residual stress in the matrix must be evaluated. And, this stress is influenced by thermal expansion mismatch between the fibers and the matrix. By assuming the same Poisson's ratio of the constituents, the residual stress in the matrix can be evaluated by [20,21]

$$\sigma_m^{R2} = E_m \frac{\lambda_2}{\lambda_1} \left[\frac{E_f}{E_L} \right] \left[\frac{V_f}{1 - \nu_m} \right] \Omega_2 \quad (9)$$

where λ_1 and λ_2 are the special coefficients involving elastic constants, they are

$$\lambda_1 = 1 - \frac{1 - 2\nu}{2 - 2\nu} (1 - E_L/E_f) \quad (10)$$

$$\lambda_2 = (1 + E_L/E_f)/2 \quad (11)$$

The misfit strain Ω_2 possesses the similar form with Ω_1 as

$$\Omega_2 = \alpha_f^I(T) - \alpha_f^I(T_0) - \alpha_m(T) + \alpha_m(T_0) \quad (12)$$

Furthermore, the radial stress at the fiber-matrix interface takes the form:

$$\sigma_1^R = -\frac{E_m}{2\lambda_1} \left[\frac{1 - \nu_f}{1 - \nu_m} \right] \Omega_2 \quad (13)$$

This stress is considered to have significant influence upon the interfacial sliding stress [24].

3.1.3. Total misfit stress

Considering the influence of porosity, thermal cracks and interfacial debonding, the distribution of TRS in the matrix and fibers is indeed complicated. In order to incorporate the internal stress release mechanisms, we present a simple solution in Fig. 6 on the basis of shear-lag assumption. For such stress diagram, the average TRS in the matrix is

$$\bar{\sigma}_m^R = (\sigma_m^{R1} + \sigma_m^{R2}) - \frac{r(\sigma_m^{R1} + \sigma_m^{R2})^2}{2\tau L_0} \quad (14)$$

where L_0 is the initial matrix crack spacing induced by TRS, r denotes the radius of carbon fibers. Note that the above equation is based on the principle of stress superposition and meanwhile

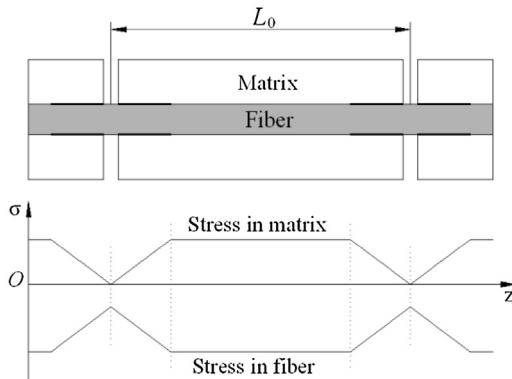


Fig. 6. Distribution of TRS in mini-composite based on shear-lag approach.

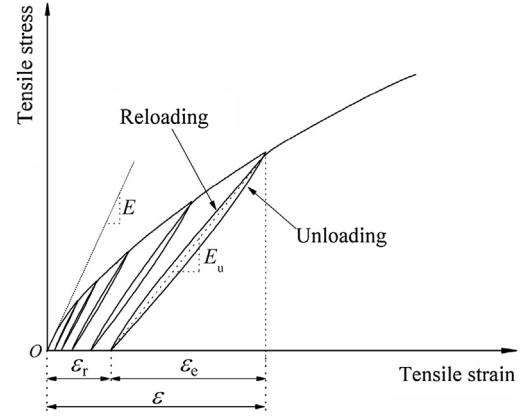


Fig. 7. Tensile strain response and strain partition of 2D-C/SiC composites.

neglects the interfacial toughness. Then, the thermal misfit stress related to the composite modulus E_c can be estimated by

$$\sigma_{th} = \frac{E_c}{E_m} \bar{\sigma}_m^R = \frac{E_L + E_T}{2E_m} \bar{\sigma}_m^R \quad (15)$$

This stress is reasonable to be utilized to predict the damage behavior such as crack opening and interfacial debonding of the longitudinal bundle.

3.2. Tensile stress-strain behavior

3.2.1. Strain partition

Based on the loading-unloading behavior in Fig. 7, the overall tensile strain upon unloading can be divided into two parts:

$$\varepsilon = \varepsilon_r + \varepsilon_e \quad (16)$$

where ε_r is the unloading residual/inelastic strain; while ε_e is the elastic strain, which has a linear form

$$\varepsilon_e = \sigma / E_u \quad (17)$$

where σ is the applied stress, E_u is the unloading modulus.

Based on the damage mechanisms analysis, the residual strain can attribute to the interfacial reverse friction upon unloading. During the previous loading process, the matrix cracks initiated and opened. However, the cracks would not be completely closed if the interfacial sliding resistance exists, and obvious irreversible deformation is retained when the sample is unloaded to stress-free state. Under such case, the magnitude of the residual strain is significantly related to the matrix crack spacing.

3.2.2. Matrix cracking model

Toward 2D-C/SiC composites, large difference of CTE exists between the constituents, which leads to high TRS inside the composites and further produces many cracks originating from the transverse layer then spreading to the longitudinal layer. Here, a new crack evolution law is suggested, which allows setting the initial and final crack spacing during an experiment. It takes the following form:

$$L = L_u + \frac{L_0 - L_u}{1 + [(\sigma + \sigma_{th}^{90} - \sigma^*) / \sigma_R]^m} \quad (18)$$

where L_u is the ultimate crack spacing; σ_R is a reference stress for matrix cracking; m is defined as the cracking exponent that is different from the classical Weibull modulus; σ^* is the minimum cracking stress. It should be noted that this model is developed from data fitting conception. Prior to application, experimental observation of the initial and final crack spacing is necessary.

3.2.3. Residual strain formulation

Based on the previous analysis, the residual strain upon unloading is mainly dominated by the crack opening displacement and crack spacing of the 0° layer, as

$$\varepsilon_r = u/L \quad (19)$$

The matrix crack opening displacement [25] can be approximated by

$$u = \Pi[2(\sigma_L + \sigma_{th}^L)^2 - 2(\sigma_D^L)^2 - \sigma_L^2] \quad (20)$$

with

$$\Pi = \frac{r(b_2 + b_3)(1 - a_1 V_f)^2}{4\tau E_m V_f^2} \quad (21)$$

where σ_L and σ_{th}^L are the applied stress and thermal misfit stress shared by the 0° layer, respectively; σ_D^L is the associated debonding stress; a_i , b_i and c_i are the special coefficients involving elastic constants defined by Hutchinson and Jensen [25].

On the other hand, the stress carried by the 90° layer σ_T can also affect the overall strain response of the composite. For simplicity, it is assumed that $\sigma_T = k\sigma_L$ with k the proportional coefficient. According to the load sharing principle, k is related to the stiffness of the 0° layer and 90° layer. Then, we obtain

$$\sigma_L = \frac{2E_L}{E_L + E_T} \sigma = \frac{2}{1+k} \sigma \quad (22)$$

Similarly, we have

$$\sigma_{th}^L = \frac{2}{1+k} \sigma_{th}, \quad \sigma_D^L = \frac{2}{1+k} \sigma_D \quad (23)$$

$$\sigma_D = \frac{1}{c_1} \sqrt{\Gamma_i E_m / r} \quad (24)$$

where Γ_i is the interface debond energy. Subsequently, the formulation of the crack opening displacement becomes

$$u = \frac{4\Pi}{(1+k)^2} [2(\sigma + \sigma_{th})^2 - 2\sigma_D^2 - \sigma^2] \quad (25)$$

Combination of Eqs. (18), (19) and (25) would give the closed form solution of the residual strain.

3.2.4. Elastic strain formulation

The mechanical parameter involved in the elastic strain is the unloading modulus, which is mainly related to the matrix cracking and interface debonding. With increasing applied load, cracks initiate in the 90° layers and rapidly spread into the 0° layers. These tunnel cracks will gradually decrease the overall stiffness of the composite. When 0° layer cracks, the interfacial debond length L_d between the fiber and matrix can be given as

$$L_d = \frac{rV_m E_m}{\tau V_f E_L (1+k)} (\sigma + \sigma_{th} - \sigma_D) \quad (26)$$

The associated instantaneous interface debonding ratio is defined as $\omega = L_d/L$. And, the post-damage modulus of the 0° layer is calculated by

$$E_L^d = \frac{V_f E_f^L E_L}{\omega E_L + (1-\omega) V_f E_f^L} \quad (27)$$

It should be noted that complete debonding reaches when $\omega=0.5$. However, this doesn't mean Eq. (27) can only be used when $\omega \leq 0.5$. Actually, it can be extended until the unloading modulus reaches fE_f based on the experimental phenomenon. Toward the post-damage modulus of the 90° layer, we still assume $E_T^d = kE_L^d$.

Hence, the overall unloading modulus of the composite can be evaluated by

$$E_u = \frac{1}{2}(E_L^d + E_T^d) = \frac{1}{2}(1+k)E_L^d \quad (28)$$

Then, the elastic strain can be obtained by application of Eqs. (17) and (28). It is worth noting that the load sharing coefficient k is virtually a variable during the continuous damage process. And, we suggest the following attenuation law

$$k = A \exp(-h/BL) + C \quad (29)$$

where A , B and C are the parameters to be determined; h is the thickness of a single layer. Eq. (29) is a mathematical approximation of the degradation phenomenon given in Ref. [26] for cross-ply laminates.

3.2.5. Total strain response

On the basis of Eqs. (16)–(19), the strain behavior under uniaxial tension can be formulated as

$$\varepsilon = u/\xi L + 2\sigma/(1+k)E_L^d \quad (30)$$

Note that an important parameter ξ is introduced here to characterize the real contribution of crack opening to the residual strain upon unloading. It can be concluded that $\xi \geq 1$ for most composites with local cracks. In general, ξ can be determined by fitting of experimental curve or by adopting empirical data. The above equation is proposed to predict the tensile stress-strain behavior of a 2D-C/SiC composite at elevated temperatures since the thermal residual stress has been taken into account.

3.3. Tensile strength

The load bearing area of the material is reduced by continuous matrix cracking. Moreover, stress concentration is introduced in the crack plane. Upon loading, cracks will propagate into weak interface. Such crack deflection other than penetration through fibers is beneficial to the failure strength. Thus, relatively low bond strength between fibers and matrix is usually designed to allow crack deflection along fiber/matrix interface.

With increasing applied stress, the overall load carrying capacity of the matrix gradually decreases. And, a lower bound will be reached when crack saturation occurs. Under such case, the maximum stress carried by matrix is $V_m \sigma_{mu}/4$ with σ_{mu} the tensile strength of matrix. The additional load will be completely suffered by the fibers. Composite rupture is assumed when the maximum stress in fibers reaches the critical value. According to our previous study [27], the average stress (within the debonding length) carried by fibers at failure can be evaluated by

$$\bar{\sigma}_f = \frac{1}{2} \left(1 + \frac{1}{K}\right) \sigma_c \quad (31)$$

where σ_c is the characteristic fiber strength parameter at the gauge length δ_c , which is twice the debonding length at stress σ_c ; while K is a stress concentration coefficient reflecting the stress distribution status in the fibers. Experiments as well as models have demonstrated that a relatively large interfacial sliding resistance is beneficial to tensile strength [28]. By contrast, too weak interfaces are shown to be detrimental. Such features are exactly incorporated into the strength parameter σ_c as:

$$\sigma_c = (\sigma_0^\rho \tau l_0 / r)^{1/(\rho+1)} \quad (32)$$

where σ_0 is the characteristic fiber strength at tested gauge length l_0 , ρ is the Weibull modulus of fiber.

Table 2

Strength data of the constituents [1,31] and the associated predicted strength of the 2D-C/SiC composite at various temperatures.

Temperature (K)	Matrix strength (MPa)	Weibull modulus	Statistical strength (MPa)	Strength (MPa)	
				Tested	Predicted
300	310	5.1	660	225.8	241.5
973	310	6.5	860	240.5	272.3
1273	310	6.5	950	268.2	294.1
1473	310	5.2	890	240.9	320.3

According to the rule of mixture, the total applied stress carried by 2D-C/SiC composites can be estimated by the following equation when neglecting the load carrying capacity of the 90° layer:

$$\sigma_u = \frac{1}{4} [V_m(1 - \phi_c)\sigma_{mu} + V_f(1 + \frac{1}{K})\sigma_c] \quad (33)$$

where ϕ_c is the critical load bearing area reduction ratio [29]. This strength model attributes the variation of strength mainly to the temperature-dependent material parameters of the constituents. Once the crack is bridged by fibers and the interface debonding arises, the stress uniformity coefficient K will remain a constant and take the following form:

$$K = 1 + (1 - V_f)E_m/(V_fE_f^L) \quad (34)$$

However, when brittle rupture occurs and the fiber pullout is quite short, K must be larger than the above. Combination of Eqs. (32)–(34) would give reasonable strength predictions.

4. Model application

4.1. Temperature-dependent parameters

The specific CTE of SiC can be expressed as [2]

$$\alpha_m(T) = -1.8276T + 0.0178T^2 - 1.5544 \times 10^{-5}T^3 + 4.5246 \times 10^{-9}T^4 \quad (35)$$

Eq. (35) is valid in the temperature range 125–1273 K. In contrast, the coefficient of thermal expansion at temperatures above 1273 K is assumed as constant $5.0 \times 10^{-6}/K$. Meanwhile, the elastic modulus of SiC with porosities at elevated temperatures can be empirically formulated as [2]

$$E_m(T) = \frac{350}{460} [460 - 0.04T \exp(-\frac{962}{T})] \quad (36)$$

Eq. (36) is reasonable in the temperature range 300–1773 K.

On the other hand, the elastic modulus of carbon fibers at elevated temperatures can be empirically expressed as [1]

$$E_f^L(T) = 230[1 - 2.86 \times 10^{-4} \exp(T/324)] \quad (37)$$

The above equation is available from ambient temperature to 2273 K. However, we considered the transverse modulus of fibers as a material constant for lack of experimental data. With respect to CTE of the fibers [30], the radial (i.e. transverse) one can be described as

$$\alpha_f^T(T) = -1.86 \times 10^{-3} + 5.85 \times 10^{-6}T - 1.36 \times 10^{-10}T^2 + 1.06 \times 10^{-14}T^3 \quad (38)$$

while the longitudinal one is

$$\alpha_f^L(T) = 2.529 \times 10^{-4} - 1.569 \times 10^{-6}T + 2.228 \times 10^{-9}T^2 - 1.877 \times 10^{-13}T^3 - 1.288 \times 10^{-16}T^4 \quad (39)$$

Eqs. (38) and (39) are valid in the temperature range 300–2500 K.

The temperature dependent strength data of carbon fibers is also required in present model. According to the experimental research

[1], the tensile strength would increase continuously when elevating the test temperature from the ambient to 1473 K. But, the growth rate is somewhat small as shown in Table 2. Tensile strength of the matrix at elevated temperature is considered as a constant for lack of experimental data, which is also listed in Table 2. The other necessary parameters presented in the analytical model are quantitatively given in Table 3.

4.2. Analysis of TRS

When cooling the 2D-C/SiC material from processing temperature 1273 K to room temperature, the interlaminar TRS calculated by Eq. (5) is about 126 MPa, which leads to cracking of transverse bundles even when no external load is applied. Meanwhile, TRS in the matrix of the longitudinal bundles derived from Eq. (14) is about 499 MPa. This stress is tensile and close to the strength of SiC

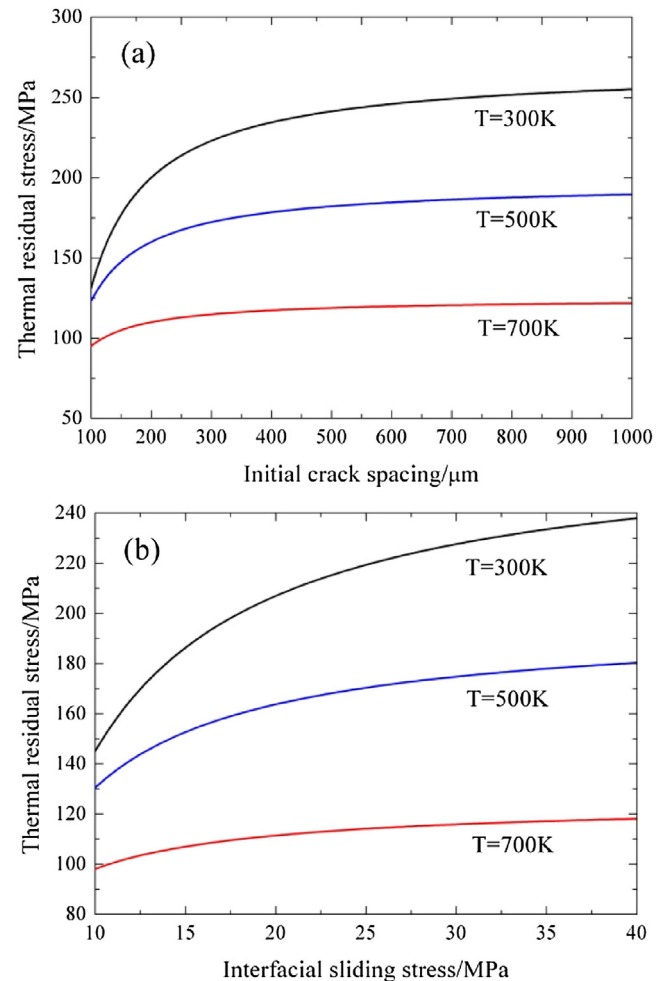


Fig. 8. Influence of (a) crack spacing and (b) interfacial sliding stress on the thermal residual stress of 2D-C/SiC material.

matrix [2]. Thus, once cracking occurred in 90° layer, it may easily spread to the adjacent 0° layer.

On the other hand, influence of matrix crack spacing and interfacial sliding upon TRS of 2D-C/SiC composite is analyzed based on Eq. (15) and schematically shown in Fig. 8. According to the model assumption and calculated results, matrix cracking and interface debonding can dramatically release the internal TRS. Moreover, for more relaxation, small crack spacing and interfacial sliding stress are beneficial.

4.3. Postulation of crack evolution

The newly proposed stochastic matrix cracking model is evaluated in this section by parametric analysis. It is seen from Fig. 9(a) that σ_R is a reference stress around which matrix cracking occurs in the absence of any thermal stresses. Fig. 9(b) presents the influence of the cracking exponent on crack evolution behavior. Actually, m is a parameter characterizing the intensity of matrix cracking around σ_R . The effect of thermal misfit stress on matrix cracking process is shown in Fig. 9(c). One can clearly see that cracking is delayed by release of thermal residual stress.

In this experiment, σ_R is identified as 30 MPa by fitting of the stress-strain curve at room temperature. Parameter σ^* is set equaling to the thermal misfit stress of 90° layer at room temperature as about 63 MPa in order to ensure that matrix cracking starts really from the onset of loading. Such phenomenon reveals different cracking process from that presented in Ref. [32]. Therefore, we

Table 3

Parameters and values used in the calculation.

Model Parameter	Symbol	value	Unit
Transverse modulus of T300 fiber	E_f^T	13.8	GPa
Interface debond energy	Γ_i	0	J/m ²
Fiber radius	r	3.5	μm
Fiber volume fraction in composite	V_f	40	%
Matrix volume fraction in composite	V_m	60	%
Initial crack spacing	L_0	1000	μm
Ultimate crack spacing	L_u	150	μm
Poisson's ratio of fiber and matrix	ν	0.2	–
Reduction factor of crack opening	ξ	1.5	–

assumed $m = 3$ to describe the cracking intensity of the 2D-C/SiC material at elevated temperatures.

4.4. Reduction of coefficient k

Variation of the load sharing coefficient k with crack spacing L is described by Eq. (29) where the parameters A , B and C must be determined. For the purpose, the boundary conditions have to be found first. It is assumed that when $L = L_0$, $k = E_T/E_L$; and when $L \rightarrow 0$, $k = 0$. As a result, we get

$$k = \frac{E_T}{E_L} \exp\left(\frac{L - 1000}{BL}\right) \quad (40)$$

Parametric analysis of B is schematically shown in Fig. 10, from which the role of B is clearly demonstrated. In order to estimate the

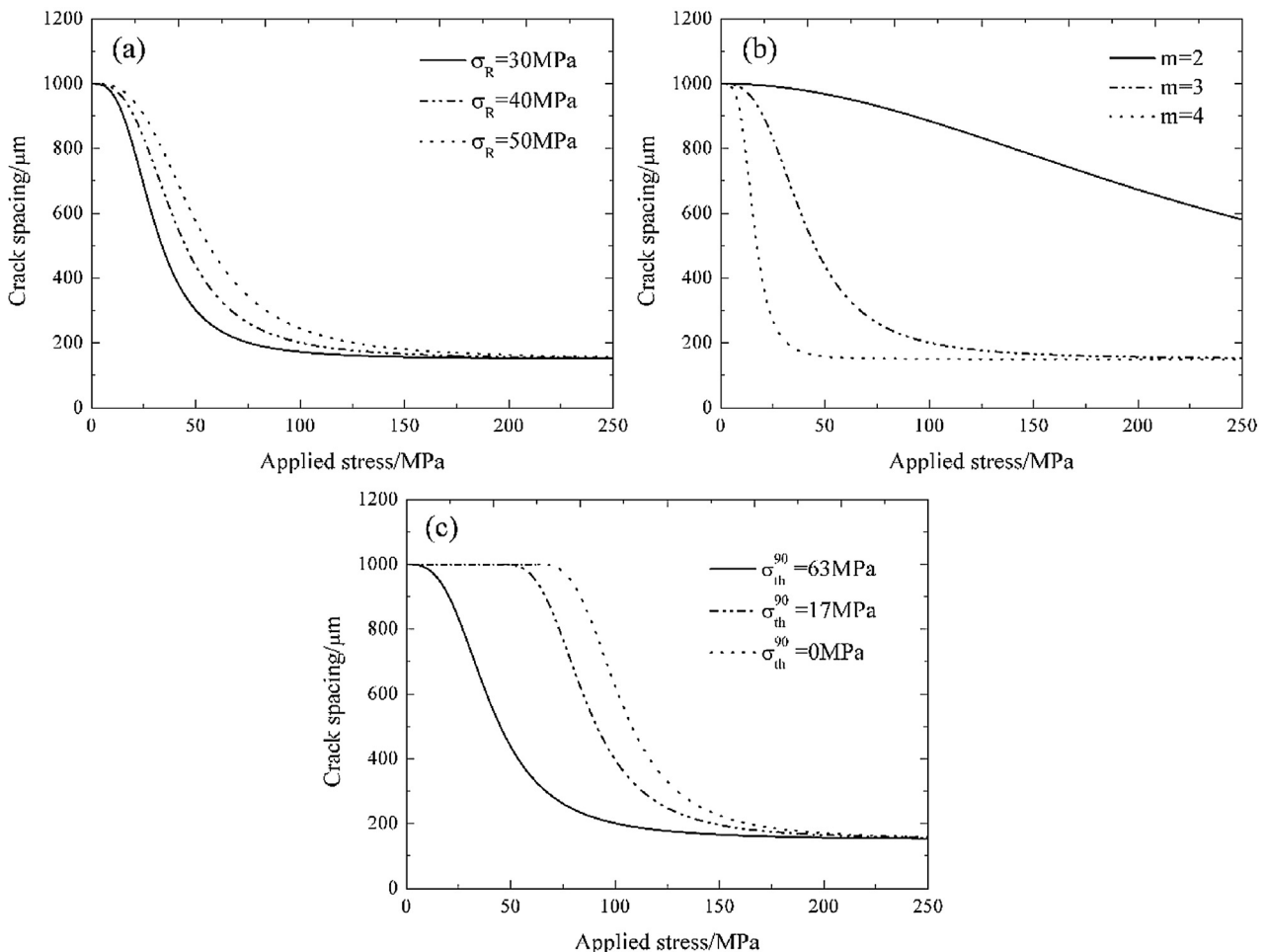


Fig. 9. Influence of (a) σ_R (with $m = 3$, $\sigma^* = \sigma_{th}^{90}$), (b) m (with $\sigma_R = 40$ MPa, $\sigma^* = \sigma_{th}^{90}$) and (c) σ_{th}^{90} (with $m = 3$, $\sigma_R = 40$ MPa, $\sigma^* = 63$ MPa) upon crack evolution of 90° layer.

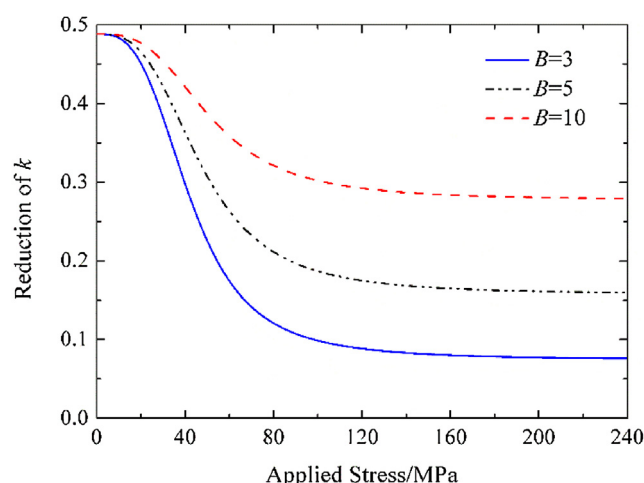


Fig. 10. Influence of B on reduction of load sharing coefficient k .

load carrying ability of the transverse fiber bundle, we prudently select $B = 5$ for 2D-C/SiC composites.

4.5. Prediction of stress-strain curves

The predicted on-axis tensile stress-strain relations for 2D-C/SiC composite at elevated temperatures are depicted in Fig. 3 with varied interfacial sliding stress. It is seen that the predicted proportional limit stress is slightly higher than the associated experimental data at high temperature. The most likely reason is that the oxidation of carbon phase has not been taken into account in the constitutive model. With no doubt, oxidation can destroy the load carrying ability of 2D-C/SiC material. On the other hand, the suggested crack evolution law may not be strictly consistent with the real cracking process. It is worth mentioning that the simulated curves with the same interfacial sliding stress at high temperatures are accurate enough. For 2D-C/SiC material, maintaining nearly stable interfacial sliding stress between 973 K and 1373 K results from the competition of thermal expansion and oxidative recession.

4.6. Prediction of tensile strength

Using the data given in Tables 2 and 3, meanwhile noticing the temperature-dependent constituent properties provided previously, the ultimate tensile strength of the 2D-C/SiC composite was theoretically calculated. The results are listed in Table 2. It is concluded that the tensile strength of 2D-C/SiC composite is dominated by fibers and increases with increasing temperature till 1273 K. In contrast with the experimental data, the predictions of Eq. (33) are slightly higher. Meanwhile, the decreasing trend of the strength beyond 1273 K was not theoretically obtained. That is because the influence of oxidation damage and thermal stress upon load-bearing mechanism of the constituents, i.e., fibers and matrix, has not been taken into account in the model.

5. Conclusions

In this paper, the uniaxial tensile behavior of a plain-weave C/SiC composite at elevated temperatures was investigated experimentally and theoretically. The tested results show the material stiffness and strength at high temperatures (973 K, 1273 K and 1473 K) are no less than that at the ambient. The temperature-dependent tensile constitutive model for plain-weave C/SiC composites is established. The model involves the influence of TRS on crack spacing, crack opening displacement and interface

debonding length. Such microscopic parameters are then correlated with the macroscopic mechanical properties by a two-scale formulation in order to describe the total stress-strain response. Moreover, tensile strength model was also developed based on rule of mixture and statistical theory, which seems irrelevant with TRS and temperature. However, the material parameters should be noticed as temperature-dependent in the strength model as well as in the constitutive model. Consequently, mechanical behavior of 2D-C/SiC composites at elevated temperatures can be easily predicted once the parameters used in the models are obtained by experiment or from material handbook. Good agreement between the model predictions and experimental data at different temperatures was then observed, which demonstrates the applicability and rationality of the models. Besides, the proposed models can also reflect well the dominated mechanisms leading to the increase of tensile modulus and strength of PAN-based carbon fiber reinforced silicon carbide matrix composites at elevated temperatures. The experimental results and theoretical models are thereby meaningful for the design of C/SiC components and structures.

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